A Review of Matrices[24]

[24] See the lecture notes from MTH1002, part 2

An $m \times n$ matrix[25] $\boldsymbol{A}$ is a rectangular array with m rows and n columns

[25] We'll use upper case bold letters for matrices and lower case bold letters for vectors. For handwritten notes, matrices are double underlined and vectors are single underlined.

$$\boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}. \tag{A.1}$$

The element of the ith row and j^{th} column is denoted by a_{ij}. The **transpose** of $\boldsymbol{A}$, denoted by $\boldsymbol{A}^T$, is obtained by interchanging the rows and columns of $\boldsymbol{A}$. The **conjugate** matrix, denoted by $\bar{\boldsymbol{A}}$, is obtained from $\boldsymbol{A}$ by replacing each element a_{ij} by its complex conjugate. The **adjoint** matrix, denoted by $\bar{\boldsymbol{A}}^T$, is the transpose of the conjugate matrix. A **square matrix** is a matrix with $m = n$. The transpose of a column vector is a row vector.

Activity A.1 Find the transpose, conjugate and adjoint of the matrix

$$A = \begin{pmatrix} 3 & 2-i \\ 4+3i & -5+2i \end{pmatrix}. \tag{A.2}$$

A.1 Properties of matrices

1. Two $m \times n$ matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are equal if $a_{ij} = b_{ij}$ for each i and j.
2. The zero matrix, denoted by $\mathbf{0}$, is the matrix in which every element is 0.
3. Addition is element-wise

$$\boldsymbol{C} = \boldsymbol{A} + \boldsymbol{B} \quad \text{means} \quad c_{ij} = a_{ij} + b_{ij}$$

and addition is commutative and associative

$$\boldsymbol{A}+\boldsymbol{B}=\boldsymbol{B}+\boldsymbol{A} \quad \text{and} \quad \boldsymbol{A}+(\boldsymbol{B}+\boldsymbol{D})=(\boldsymbol{A}+\boldsymbol{B})+\boldsymbol{D}\,.$$

4. Multiplication by a number is element-wise

$$\boldsymbol{C}=\alpha\boldsymbol{A} \quad \text{means} \quad c_{ij}=\alpha\, a_{ij}$$

and multiplication by a number is distributive

$$\alpha(\boldsymbol{A}+\boldsymbol{B})=\alpha\boldsymbol{A}+\alpha\boldsymbol{B} \quad \text{and} \quad (\alpha+\beta)\boldsymbol{A}=\alpha\boldsymbol{A}+\beta\boldsymbol{A}\,.$$

5. Matrix multiplication (for example, the product $\boldsymbol{AB}$) is defined whenever the number of columns of the first matrix is the same as the number of rows of the second matrix. For example, if $\boldsymbol{A}$ is an $m\times n$ matrix and $\boldsymbol{B}$ is an $n\times r$ matrix then $\boldsymbol{C}=\boldsymbol{AB}$ is an $m\times r$ matrix with

$$c_{ij}=\sum_{k=1}^{n} a_{ik}b_{kj}\,.$$

Matrix multiplication is associative and distributive

$$(\boldsymbol{AB})\boldsymbol{C}=\boldsymbol{A}(\boldsymbol{BC}), \quad \text{and} \quad \boldsymbol{A}(\boldsymbol{B}+\boldsymbol{C})=\boldsymbol{AB}+\boldsymbol{AC}$$

but in general it is not commutative, i.e.

$$\boldsymbol{AB}\neq\boldsymbol{BA}\,.$$

6. As a special case, the multiplication of a $1\times n$ row vector and an $n\times 1$ column vector, denoted by $\boldsymbol{x}^T$ and $\boldsymbol{y}$ respectively, is

$$\boldsymbol{x}^T\boldsymbol{y}=\sum_{i=1}^{n} x_i y_i$$

and the **scalar (or inner) product** of $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined by

$$(\boldsymbol{x},\boldsymbol{y})=\boldsymbol{x}^T\bar{\boldsymbol{y}}=\sum_{i=1}^{n} x_i\bar{y}_i\,,$$

recalling that a bar means complex conjugate.

The scalar product of $\boldsymbol{x}$ with itself gives a non-negative real number

$$(\boldsymbol{x},\boldsymbol{x})=\sum_{i=1}^{n} x_i\bar{x}_i=\sum_{i=1}^{n}|x_i|^2$$

called the **magnitude** of $\boldsymbol{x}$.

If $(\boldsymbol{x},\boldsymbol{y})=0$ then the two vectors are said to be **orthogonal** (e.g. the basis vectors for Cartesian geometry).

7. The **identity matrix**

$$\boldsymbol{I}=\begin{pmatrix} 1 & 0 & \dots & 0\\ 0 & 1 & \dots & 0\\ \vdots & \vdots & & \vdots\\ 0 & 0 & \dots & 1 \end{pmatrix} \tag{A.3}$$

and

$$\boldsymbol{AI}=\boldsymbol{IA}=\boldsymbol{A}\,.$$

8. If there are two square matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ with $\boldsymbol{AB} = \boldsymbol{I}$ then $\boldsymbol{B}$ is the **inverse** of $\boldsymbol{A}$ and we write

$$\boldsymbol{B} = \boldsymbol{A}^{-1}.$$

If $\boldsymbol{A}^{-1}$ exists then $\boldsymbol{A}$ is said to be nonsingular and we have

$$\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}.$$

If $\boldsymbol{A}^{-1}$ does not exist then $\boldsymbol{A}$ is said to be **singular**.

Assuming that $\boldsymbol{A}^{-1}$ exists, there are various ways to compute it. One way is to use determinants:

- Find the **minor** M_{ij} by calculating the determinant of the matrix which is obtained from $\boldsymbol{A}$ by deleting the ith row and the jth column (i.e. the row and column containing a_{ij}).
- Form the **cofactor**
$$C_{ij} = (-1)^{i+j} M_{ij}.$$
- The inverse $\boldsymbol{B} = \boldsymbol{A}^{-1}$ has element
$$b_{ij} = \frac{1}{|\boldsymbol{A}|} C_{ji},$$
where $|\boldsymbol{A}|$ is the determinant of $\boldsymbol{A}$. If $|\boldsymbol{A}| = 0$ then $\boldsymbol{A}$ is singular.

For large n a more efficient way is to use **row reduction** to transform $\boldsymbol{A}$ into $\boldsymbol{I}$. The same series of row operations then transforms $\boldsymbol{I}$ into $\boldsymbol{A}^{-1}$. The three types row operations are:

- Interchanging two rows.
- Multiplying any row by a non-zero scalar.
- Adding a multiple of one row to another row.

Activity A.2 Find the inverse of

$$A = \begin{pmatrix} 1 & -1 & -1 \\ 3 & -1 & 2 \\ 2 & 2 & 3 \end{pmatrix}. \tag{A.4}$$

Make sure that you check your answer by computing AA^{-1}.

9. We will consider matrix functions, i.e. matrices whose elements are functions of a real variable (t, say), for example the $n \times n$ matrix function

$$\boldsymbol{A}(t) = \begin{pmatrix} a_{11}(t) & a_{12}(t) & \dots & a_{1n}(t) \\ a_{21}(t) & a_{22}(t) & \dots & a_{2n}(t) \\ \vdots & \vdots & & \vdots \\ a_{n1}(t) & a_{n2}(t) & \dots & a_{nn}(t) \end{pmatrix}. \tag{A.5}$$

The matrix is said to be continuous at $t = t_0$ (or on an interval $I : \alpha < t < \beta$) if each element of $\boldsymbol{A}$ is a continuous function at t_0 (or on the interval I). Similarly, $\boldsymbol{A}$ is said to be differentiable if each of its elements are differentiable, and the derivative $\frac{d\boldsymbol{A}}{dt}$ is the matrix obtained by differentiating each of the elements a_{ij}. Similarly, the integral of a matrix function is the matrix obtained by integrating each of the elements a_{ij}.

Example A.3 Consider the matrix function

$$\boldsymbol{A} = \begin{pmatrix} \sin t & t \\ 1 & \cos t \end{pmatrix}. \tag{A.6}$$

Find

$$\frac{d\boldsymbol{A}}{dt} \quad \text{and} \quad \int_0^{\pi} \boldsymbol{A}(t)dt\,.$$